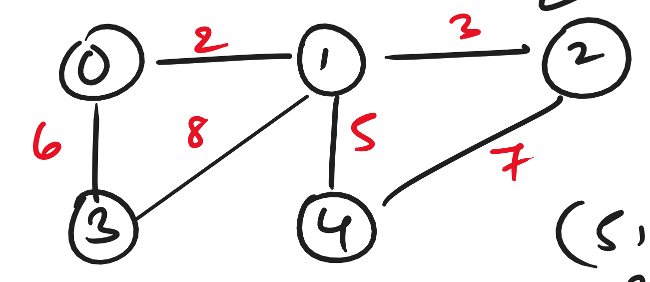
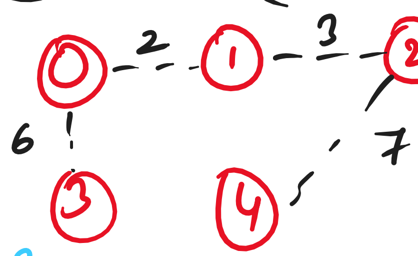
Minimum Spanning Tree :→ MST



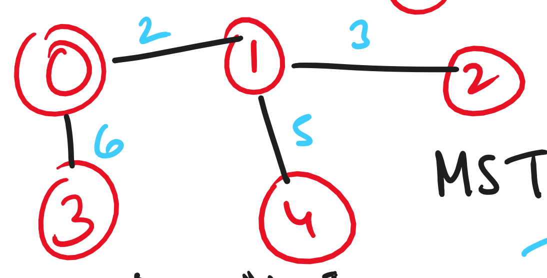
Not the (5,4) least

ST ⇒



Spanning Tree Defn

A tree in which there are N nodes & N-1 edges is called a Spanning Tree, where all nodes are reachable from every other node.

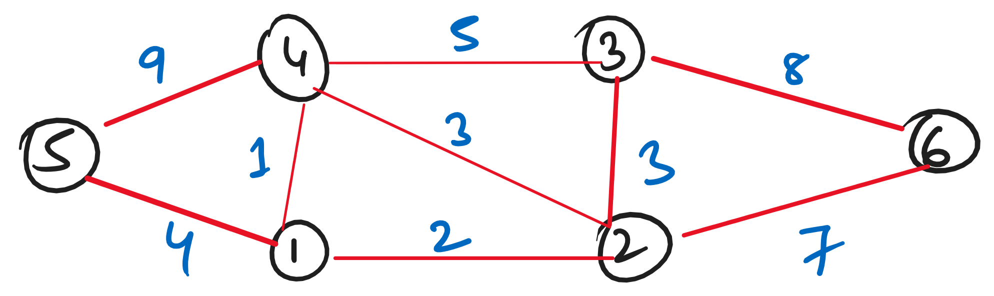


Spanning Tree Edge Sum = 6+2+3+7 = 18

MST sum = 16

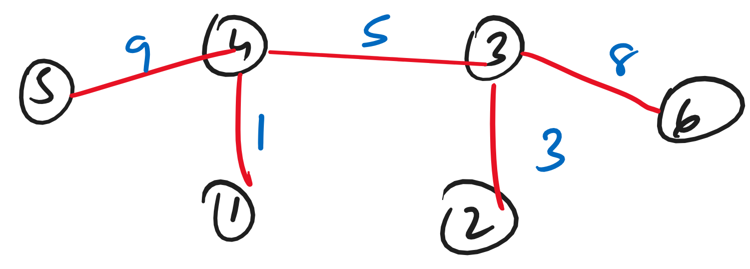
The tree having the least edge sum is the Minimum Spanning Tree.

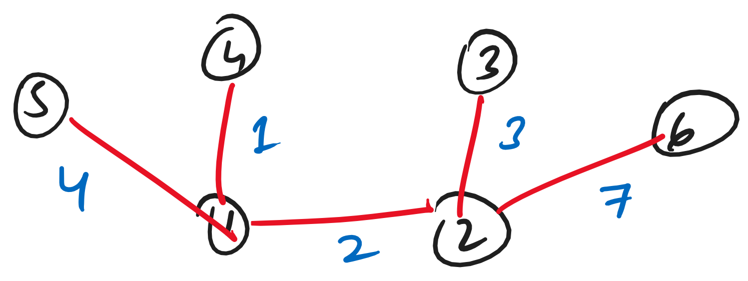
Draw the MST for the given graph :→



6 Nodes
6-1 = 5 edges

Sum
MST – 17

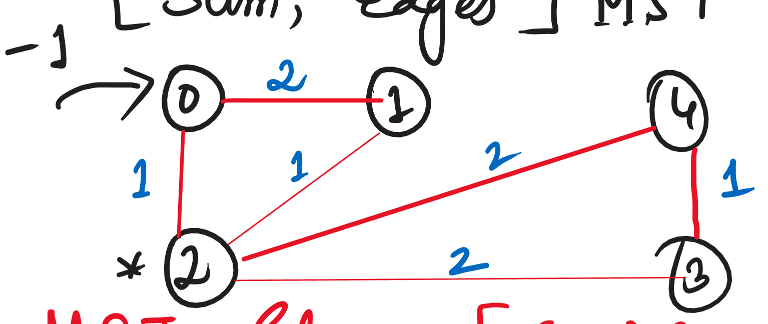




Prim's Algorithm for Minimum Spanning Tree

[Sum, Edges] MST

Sum = [0+1+1+2+1] = 5



* only mark vis after reaching

wt,	n,	p
0	0,	0, -1
1	1,	2, 0
1	1,	1, 2
~~2~~	2,	1, 0
2	2,	3, 2
1	1,	4, 3

×2, 2, 4, 2

empty

~~(1, 4, 3)~~
~~2, 3, 2~~
~~2, 4, 2~~
~~1, 1, 2~~
~~(1, 2, 0)~~
~~(2, 1, 0)~~
~~(0, 0, -1)~~

{wt, node, parent}

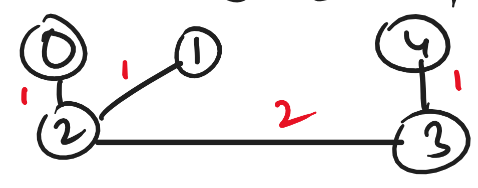
pq → min heap

MST - Edges = [(0,1,2) (1,1,2) (2,1,3) (3,1,4)]

visited[] =

0	1	2	3	4
~~0~~ 1	~~0~~ 1	~~0~~ 1	~~0~~ 1	~~0~~ 1

Find MST Diagram :→



= 5

Important Graph Algos & Analogies for Interviews (Coding + Tech)

(i) Representation : Matrix & List (Adjacency)
(ii) Traversals : DFS (Recursion) BFS (Queue)
(iii) Cycle Detection using BFS / DFS
(iv) Linear Ordering → Topological Sort
(v) Topo Sort DFS / BFS
(vi) Topo Sort Kahn's Algorithm
(vii) Topo Sort Kahn's Algorithm (size == N)
(viii) Shortest Distance Algorithms:
Single src ∈ (a) Dijkstra's (+) ve weights
* (b) Bellman Ford's (+) ve & (-) ve
* (c) Floyd Warshall's (+) ve & (-) ve
(ix) Minimum Spanning Tree
* Prim's Algo
* Kruskal's Algo (4α)
{ Disjoint Set Data Structure }
(1) Union < size, rank
(2) Find Ultimate Parent
(x) Kosaraju's Algorithm (Strongly Connected Components) (SCC)
(xi) Tarjan's Algorithm

DRY RUN

(i) Count edges & nodes in a graph
(ii) Convert adj matrix to adj list
(iii) Number of Islands } GFG
(iv) Number of Provinces
(v) Viva → Dijkstra's Problem
(vi) Time Complexities (MCQ / Viva)
(450dsa.com) 450 dsa Questions